

SIGNEX — SIGNAL EXTRACTION & ANALYSIS ENGINE

NTRO ID26147 CLASSIFIED INTELLIGENCE DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: e4ec1ed512d3bbf2

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : sample_qpsk.iq
File SHA-256 Digest : e4ec1ed512d3bbf224bdf0603f5e0aca5a82e8831bfe6a32a87a6d8a872d3858
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-19T08:06:14Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	2.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+0.923 kHz	Spectral Centroid Offset
Estimated Baud Rate	500.00 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	976.6 Hz	Half-Power Frequency Span
Occupied Bandwidth (99%)	1706.05 kHz	Cumulative PSD 99% Energy
Estimated SNR	12.31 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	3.58 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+0.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.1365	Wiener Entropy Index (0.0 – 1.0)

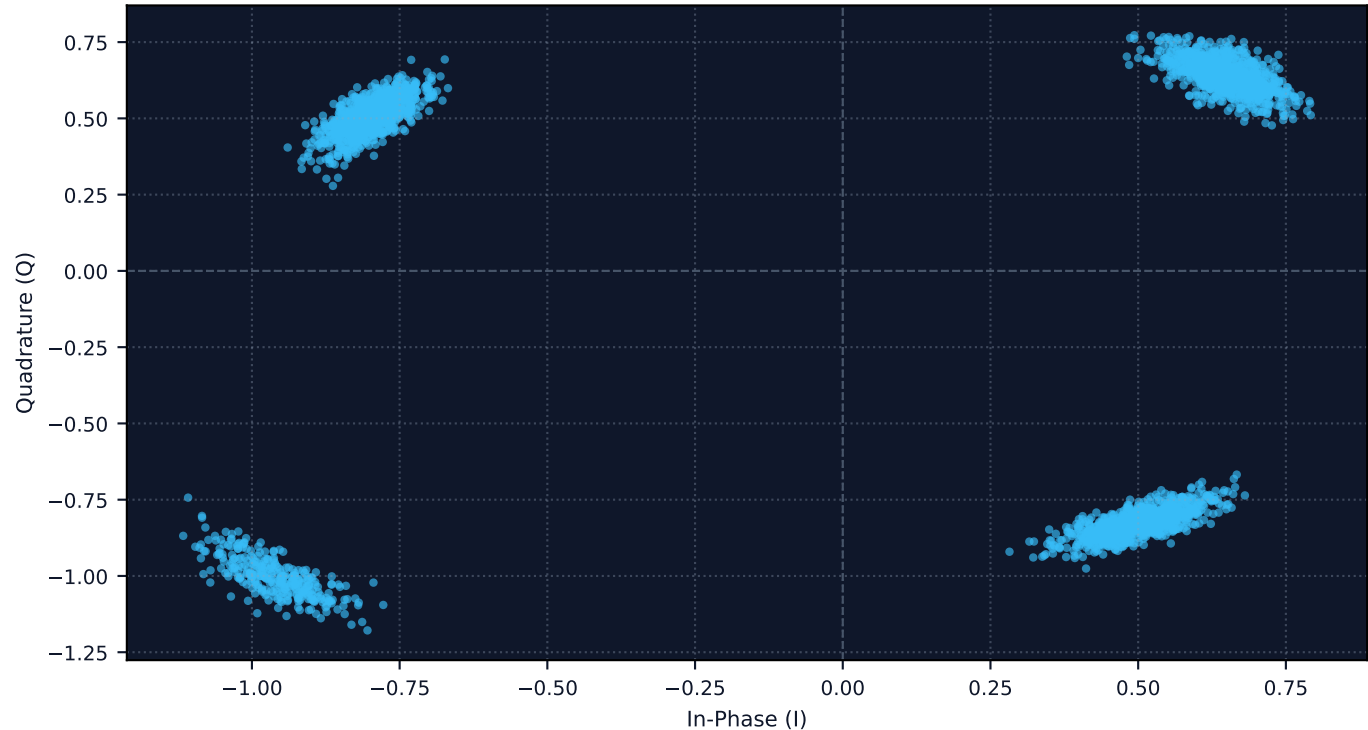
PREDICTED MODULATION : WBFM

CLASSIFIER CONFIDENCE : 98.06% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: QPSK | Recovered Bits: 16,384 | Estimated EVM: 10.01% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [WBFM]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	196	0x0000	0	2016 B	CRC MISMATCH	PARTIAL_RECOVER
#2	258	0x0000	1	2009 B	CRC MISMATCH	PARTIAL_RECOVER
#3	552	0x0000	2	1972 B	CRC MISMATCH	PARTIAL_RECOVER
#4	628	0x0000	3	1962 B	CRC MISMATCH	PARTIAL_RECOVER
#5	1353	0x0000	4	1872 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 FCF53AD5B5F4D2D35BFA9293FD5DD4D5 [FRAME 1 PAYLOAD]